

STAT 9120: Theory of Linear Models

Homework 2: Estimability

Instructions. Show all work and justify your answers. For proof-based questions, clearly state the facts or results you use. Unless otherwise stated, consider the general linear model

$$Y = X\beta + \varepsilon, \quad E(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I,$$

where $X \in \mathbb{R}^{n \times p}$ is not necessarily of full column rank.

Recall that a linear parametric function $c^T \beta$ is **estimable** if there exists a linear statistic $a^T Y$ such that

$$E(a^T Y) = c^T \beta$$

for every β .

Problem 1. The Basic Criterion for Estimability

[16 points]

Let $c \in \mathbb{R}^p$.

- (a) Show that $c^T \beta$ is estimable if and only if there exists $a \in \mathbb{R}^n$ such that

$$X^T a = c.$$

[7 points]

- (b) Use part (a) to show that

$$c^T \beta \text{ is estimable} \iff c \in \mathcal{C}(X^T).$$

[4 points]

- (c) Explain in words what this result says about which linear combinations of the entries of β can be learned from the data.

[5 points]

Problem 2. Estimability and the Null Space

[16 points]

Show that the following statements are equivalent:

1. $c^T \beta$ is estimable;
2. $c \in \mathcal{C}(X^T)$;
3. $c \in \mathcal{N}(X)^\perp$;
4. for every $h \in \mathcal{N}(X)$, $c^T h = 0$.

Then explain why an estimable function must have the same value for any two parameter vectors β_1 and β_2 satisfying

$$X\beta_1 = X\beta_2.$$

Interpret this result statistically.

[16 points]

Problem 3. Determining Estimable Functions**[18 points]**

Consider the design matrix

$$X = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{pmatrix}.$$

(a) Find a nonzero vector h such that $Xh = 0$. Hence determine $\text{rank}(X)$. **[4 points]**

(b) For each of the following functions, determine whether it is estimable. Justify each answer.

- i. β_1 , ii. β_2 , iii. β_3 ,
 iv. $\beta_1 + \beta_3$, v. $\beta_2 + \beta_3$, vi. $\beta_1 - \beta_2 + \beta_3$.

[9 points](c) Find a basis for the set of all $c \in \mathbb{R}^3$ for which $c^T\beta$ is estimable. **[3 points]**(d) How many linearly independent estimable functions of β are there? Relate your answer to $\text{rank}(X)$. **[2 points]****Problem 4. A Rank-Deficient Effects Model****[16 points]**

Suppose

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{pmatrix}.$$

Thus the model can be written as

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2.$$

(a) Show that X is rank deficient and find a nonzero vector in $\mathcal{N}(X)$. **[4 points]**(b) Show that neither μ , α_1 , nor α_2 is individually estimable. **[4 points]**

(c) Determine whether each of the following is estimable:

$$\mu + \alpha_1, \quad \mu + \alpha_2, \quad \alpha_1 - \alpha_2.$$

Justify your answers.

[5 points](d) Interpret the estimable quantities from part (c) in the context of this model. **[3 points]**

Problem 5. Estimability and Generalized Inverses**[18 points]**

Let

$$A = X^T X$$

and let G be any generalized inverse of A , so that

$$AGA = A.$$

Define

$$\hat{\beta}_G = GX^T Y.$$

- (a) Show that
- $\hat{\beta}_G$
- satisfies the normal equations

$$X^T X \hat{\beta} = X^T Y.$$

You may use the result

$$\mathcal{C}(X^T X) = \mathcal{C}(X^T).$$

[6 points]

- (b) Suppose
- $c^T \beta$
- is estimable. Show that
- $c^T \hat{\beta}$
- has the same value for every solution
- $\hat{\beta}$
- of the normal equations.
- [6 points]**

- (c) Explain why
- $\hat{\beta}_G$
- itself can depend on the generalized inverse
- G
- , while the estimate of an estimable function
- $c^T \hat{\beta}_G$
- does not.
- [3 points]**

- (d) Explain the distinction between the statements “
- $\hat{\beta}$
- is unique” and “
- $X\hat{\beta}$
- is unique.” Which statement remains true in a rank-deficient linear model?
- [3 points]**

Problem 6. Fitted Values in a Rank-Deficient Model**[16 points]**Suppose B is a generalized inverse of $X^T X$, and define

$$P = XBX^T.$$

- (a) Show that

$$XB(X^T X) = X.$$

Hint: Use the generalized-inverse identity and the fact that $\mathcal{N}(X^T X) = \mathcal{N}(X)$.**[5 points]**

- (b) Use part (a) to show that

$$P^2 = P.$$

[4 points]

- (c) Show that
- P
- acts as the identity on
- $\mathcal{C}(X)$
- ; that is,

$$Py = y \quad \text{for every } y \in \mathcal{C}(X).$$

[4 points]

- (d) Explain why the fitted value

$$\hat{Y} = X\hat{\beta}$$

is unique even when the least-squares estimate $\hat{\beta}$ is not unique.**[3 points]**

Total: 100 points

Key Ideas to Review

The central characterizations of estimability are

$$c^T \beta \text{ estimable} \iff c \in \mathcal{C}(X^T)$$

and

$$\mathcal{C}(X^T) = \mathcal{N}(X)^\perp.$$

Thus,

$$c^T \beta \text{ estimable} \iff c^T h = 0 \text{ for every } h \in \mathcal{N}(X).$$

For a rank-deficient model, the parameter vector β need not be identifiable, but the mean vector

$$X\beta$$

is identifiable. Similarly, least-squares solutions $\hat{\beta}$ need not be unique, but

$$X\hat{\beta}$$

is unique.

A useful way to think about estimability is:

An estimable function is a function of β that does not change when β is moved in a direction that leaves the mean vector $X\beta$ unchanged.